

Discrete Mathematics
MATH1031 - Durham University

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1 Introduction to Discrete Mathematics

1.1 Arrangements and Permutations

1.1.1 Arrangements

Definition 1.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 1.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1.1.1.1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 1.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

1.2 Combinations

1.3 Arrangements and Combinations with Repetitions

2 Combinatorics

2.1 The Three Principles

2.2 Recurrence Relations

2.3 Generating Functions

3 Graph Theory

3.1 What is a Graph?

3.2 Eulerian Graphs

3.3 Planar Graphs